

Simplicial Methods in Algebra

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1 Simplicial Homological Algebra

1.1 Chain Complexes of Simplicial Objects

Definition 1.1.1 (Associated Chain Complex) Let $\mathcal{A}$ be an abelian category. Let A be a (semi)-simplicial object in $\mathcal{A}$. Define the associated chain complex of A to be

$$\cdots \longrightarrow C_{n+1}(A) \xrightarrow{\partial_{n+1}} C_n(A) \xrightarrow{\partial_n} C_{n-1}(A) \longrightarrow \cdots \longrightarrow C_0(A)$$

where $C_n(A) = A_n$ and the boundary operator given by

$$\partial_n = \sum_{i=0}^n (-1)^i d_i^n : A_n \rightarrow A_{n-1}$$

TBA: Functoriality of associated chain complex

Definition 1.1.2 (Simplicial Homology) Let R be a ring. Let X be a (semi)-simplicial set. Define the simplicial homology of X with coefficients in R to be the homology groups

$$H_n^\Delta(X; R) = H_n(C_\bullet(R[X]))$$

Notice that this definition coincides with that in Algebraic Topology 2. Recall that in AT2 we defined the simplicial homology of a Δ -set, but in $\mathbb{Z}$ coefficients.

1.2 Normalized Chain Complexes

Definition 1.2.1 (Normalized Chain Complexes) Let $\mathcal{A}$ be an abelian category or the category $\mathbf{Grp}$. Let A be a simplicial object in $\mathcal{A}$. Define the normalized chain complex of A to be the chain complex:

$$\cdots \longrightarrow N_{k+1}(A) \xrightarrow{\partial_{k+1}} N_k(A) \xrightarrow{\partial_k} N_{k-1}(A) \longrightarrow \cdots \longrightarrow N_1(A)$$

where

$$N_k(A) = \bigcap_{i=1}^k \ker(d_i^k : A_k \rightarrow A_{k-1})$$

and the differential given by $\partial_k = d_0^K|_{N_k(A)}$. We denote the normalized chain complex by $(N_\bullet(G), \partial_\bullet)$

nLab: We may think of the elements of the complex in degree k as being k -dimensional disks in G all of whose boundary is captured by a single face.

Lemma 1.2.2 Let G be a simplicial group. Consider the normalized chain complex $(N_\bullet(G), \partial_\bullet)$. Then $\partial_n N_n(G)$ is a normal subgroup of $N_{n-1}(G)$.

Because of this lemma, it now makes sense to take the homology group of the normalized chain complex even if we take a simplicial object in $\mathbf{Grp}$.

Definition 1.2.3 (Normalized Chain Complex Functor) Let $\mathcal{A}$ be an abelian category. Define the normalized chain complex functor N

Definition 1.2.4 (Degenerate Chain Complex) Let $\mathcal{A}$ be an abelian category. Let A be a simplicial object in $\mathcal{A}$. Define the degenerate chain complex $D_\bullet(A)$ of A to be the subcomplex of the associated chain complex $C_\bullet(A)$ defined by

$$D_n(A) = \langle s_i^n : A_n \rightarrow A_{n+1} \mid s_i \text{ is the degenerate maps} \rangle$$

Proposition 1.2.5 Let $\mathcal{A}$ be an abelian category. Let A be a simplicial object in $\mathcal{A}$. Then there is a splitting

$$C_{\bullet}(A) \cong N_{\bullet}(A) \oplus D_{\bullet}(A)$$

in the abelian category of chain complexes of $\mathcal{A}$.

Theorem 1.2.6 (Eilenberg-MacLane) Let $\mathcal{A}$ be an abelian category. Let A be a simplicial object in $\mathcal{A}$. Then the inclusion

$$N_{\bullet}(A) \hookrightarrow C_{\bullet}(A)$$

is a natural chain homotopy equivalence. In other words, $D_{\bullet}(A)$ is null homotopic.

Theorem 1.2.7 (The Dold-Kan Correspondence) Consider the abelian category $\mathbf{Ab}$ of abelian groups. The normalized chain complex functor

$$N : \mathbf{sAb} \xrightarrow{\cong} \mathbf{Ch}_{\geq 0}(\mathbf{Ab})$$

gives an equivalence of categories, with inverse as the simplicialization functor

$$\Gamma : \mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \rightarrow \mathbf{sAb}$$

1.3 Bar Resolutions

Definition 1.3.1 (Bar Construction)

Let R be a ring. Let A be an algebra over R . Let M be an A -algebra. Define the maps $d_i^n : M \otimes A^{\otimes n} \rightarrow M \otimes A^{\otimes n-1}$ by the following formulas:

- If $i = 0$, then

$$d_i^n(m \otimes a_1 \otimes \cdots \otimes a_n) = ma_1 \otimes a_2 \otimes \cdots \otimes a_n$$

- If $0 < i < n$, then

$$d_i^n(m \otimes a_1 \otimes \cdots \otimes a_n) = m \otimes a_1 \otimes \cdots \otimes a_{i-1} \otimes a_i a_{i+1} \otimes a_{i+2} \otimes \cdots \otimes a_n$$

- If $i = n$, then

$$d_i^n(m \otimes a_1 \otimes \cdots \otimes a_n) = ma_n \otimes a_1 \otimes a_2 \otimes \cdots \otimes a_{n-1}$$

Proposition 1.3.2 Let A be an algebra over a ring R . Let M be an A -algebra. Then $(M \otimes A^{\otimes n}, d_i^n)$ defines a simplicial object in ????

Definition 1.3.3 (Bar Resolutions) Let A be an algebra over a ring R . Let M be an A -algebra. Define the bar resolution of M to be the associated chain complex of the simplicial object

$$(M \otimes A^{\otimes n}, d_i^n)$$

Explicitly, the chain complex is given in the form

$$\cdots \longrightarrow A^{\otimes n+1} \otimes M \longrightarrow A^{\otimes n} \otimes M \longrightarrow A^{\otimes n-1} \otimes M \longrightarrow \cdots \longrightarrow A \otimes M \longrightarrow M \longrightarrow 0$$

with the boundary map $\partial : A^{\otimes n} \otimes M \rightarrow A^{\otimes n-1} \otimes M$ given by

$$\partial = \sum_{i=0}^n (-1)^i d_i^n$$

2 (Co)Homology of Groups

2.1 Invariants and Coinvariants

Definition 2.1.1 (The Group of Invariants)

Let G be a group and let M be a G -module. Define the group of invariants of G in M to be the subgroup

$$M^G = \{m \in M \mid gm = m \text{ for all } g \in G\}$$

This is the largest subgroup of M for which G acts trivially.

Definition 2.1.2 (Functor of Invariants)

Let G be a group. Define the functor of invariants by

$$(-)^G : {}_G\mathbf{Mod} \rightarrow \mathbf{Ab}$$

as follows.

- For each G -module M , M^G is the group of invariants
- For each morphism $f : M \rightarrow N$ of G -modules, $f^G : M^G \rightarrow N^G$ is the restriction of f to M^G .

Proposition 2.1.3

Let G be a group. The functor of invariants $(-)^G : {}_G\mathbf{Mod} \rightarrow \mathbf{Ab}$ is left exact.

Definition 2.1.4 (The Group of Coinvariants)

Let G be a group and let M be a G -module. Define the group of coinvariants of G in M to be the quotient group

$$M_G = \frac{M}{\langle gm - m \mid g \in G, m \in M \rangle}$$

This is the largest quotient of M for which G acts trivially.

2.2 Group Cohomology and its Equivalent Forms

Definition 2.2.1 (The n th Cohomology Group) Let G be a group. Define the n th cohomology group of G with coefficients in a G -module M to be

$$H^n(G; M) = (R^n(-)^G)(M)$$

the n th right derived functor of $(-)^G : {}_G\mathbf{Mod} \rightarrow \mathbf{Ab}$.

Theorem 2.2.2 Let G be a group and let M be a G -module. Then there is an isomorphism

$$H^n(G; M) \cong \mathrm{Ext}_{\mathbb{Z}[G]}^n(\mathbb{Z}, M)$$

that is natural in M .

Recall that there are two descriptions of Ext by considering it as a functor of the first or second variable. Since the above theorem exhibits an isomorphism that is natural in the second variable, let us consider Ext as the right derived functor of the functor $\mathrm{Hom}_{\mathbb{Z}[G]}(-, M)$ applied to $\mathbb{Z}$ as a $\mathbb{Z}[G]$ -module.

Proposition 2.2.3 Let G be a group and let M be a G -module. Let $P_\bullet \rightarrow \mathbb{Z}$ be a projective resolution of $\mathbb{Z}$ with $\mathbb{Z}[G]$ -modules. Then there is an isomorphism

$$H^n(G; M) \cong H^n(\mathrm{Hom}_{\mathbb{Z}[G]}(P_\bullet, M))$$

that is natural in M .

For any group G , there is always the trivial choice of projective resolution. In the following lemma, we

use the notation $(g_0, \dots, \hat{g}_i, \dots, g_n)$ as a shorthand for writing the element in G^n but with the i th term omitted.

Lemma 2.2.4 Let G be a group. Then the cochain complex

$$\cdots \longrightarrow \mathbb{Z}[G^{n+1}] \xrightarrow{f_n} \mathbb{Z}[G^n] \xrightarrow{f_{n-1}} \mathbb{Z}[G^{n-1}] \longrightarrow \cdots \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z} \longrightarrow 0$$

where $f_n : \mathbb{Z}[G^{n+1}] \rightarrow \mathbb{Z}[G^n]$ is defined by

$$(g_0, \dots, g_n) \mapsto \sum_{i=0}^n (-1)^i (g_0, \dots, \hat{g}_i, \dots, g_n)$$

is a projective resolution of $\mathbb{Z}$ lying in ${}_{\mathbb{Z}[G]}\mathbf{Mod}$.

Let A be an R -algebra and let M be an A -module. Recall that the bar resolution is defined to be the chain complex consisting of $M \otimes A^{\otimes n}$ for each $n \in \mathbb{N}$ together with the boundary maps defined by multiplying the i th element to the $i + 1$ th element. Now let G be a group. By considering $\mathbb{Z}[G]$ as a $\mathbb{Z}$ -algebra and that and ring is a module over itself, it makes sense to talk about the bar resolution of $\mathbb{Z}[G]$.

Proposition 2.2.5 Let G be a group. Consider the bar resolution

$$\cdots \longrightarrow \mathbb{Z}[G^{n+1}] \longrightarrow \mathbb{Z}[G^n] \longrightarrow \mathbb{Z}[G^{n-1}] \longrightarrow \cdots \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z} \longrightarrow 0$$

of $\mathbb{Z}[G]$. Then it is a free resolution, and hence a projective resolution of $\mathbb{Z}$ with $\mathbb{Z}[G]$ -modules.

Thus, given a group G and a G -module M , the group cohomology of G with coefficients in M can be thought of in the following way:

- It is the right derived functor of the functor of invariants $(-)^G : {}_G\mathbf{Mod} \rightarrow \mathbf{Ab}$
- It is the extension group $\mathrm{Ext}_{\mathbb{Z}[G]}^n(\mathbb{Z}, M)$ (which is computable by the obvious projective resolution $\mathbb{Z}[G^\bullet]$)

2.3 Group Homology and its Equivalent Forms

Definition 2.3.1 (The n th Homology Group) Let G be a group. Define the n th homology group of G with coefficients in a G -module M to be

$$H_n(G; M) = (L_n(-)_G)(M)$$

the n th left derived functor of $(-)_G : {}_G\mathbf{Mod} \rightarrow \mathbf{Ab}$.

Theorem 2.3.2 Let G be a group and let M be a G -module. Then there is an isomorphism

$$H_n(G; M) \cong \mathrm{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

that is natural in M .

2.4 Low Degree Interpretations

Theorem 2.4.1 Let G be a group and let M be a G -module. Then there are natural isomorphisms

$$H^0(G, M) = M^G \quad \text{and} \quad H_0(G; M) = M_G$$

Theorem 2.4.2 Let G be a group and let M be a G -module. Then there is an isomorphism

$$H_1(G, M) \cong \frac{G}{[G, G]} = G_{\text{ab}}$$

Theorem 2.4.3 Let G be a group and let M be a trivial G -module. Then there is a natural isomorphism

$$H^1(G; M) = \frac{(\{f : G \rightarrow M \mid f(ab) = f(a) + af(b)\}, +)}{\langle f : G \rightarrow M \mid f(g) = gm - m \text{ for some fixed } m \rangle}$$

Corollary 2.4.4 Let G be a group and let M be a trivial G -module. Then there is a natural isomorphism

$$H^1(G; M) \cong \text{Hom}_{\mathbf{Grp}}(G, M)$$

3 Hochschild (Co)Homology for Rings

3.1 The Enveloping Algebra

Definition 3.1.1 (Enveloping Algebra) Let A be an R -algebra. Define the enveloping algebra of A to be

$$A^e = A \otimes A^{\text{op}}$$

Proposition 3.1.2 Let A be an R -algebra. Then any (A, A) -bimodule M equal to a left (right) A^e -module.

3.2 The Hochschild Complex

Definition 3.2.1 (The Hochschild Complex)

Let R be a commutative ring. Let A be an R -algebra. Let M be an (A, A) -bimodule. Define the Hochschild complex

$$C_*(A, M)$$

to be the associated chain complex of the simplicial object defined as follows: For each $n \in \mathbb{N}$, the n th term is given by

$$M \otimes_R A^{\otimes_R^n}$$

The face maps $d : C_n(A, M) \rightarrow C_{n-1}(A, M)$ is given by the following formula:

- $d_0(m \otimes a_1 \otimes \cdots \otimes a_n) = ma_1 \otimes a_2 \otimes \cdots \otimes a_n$
- $d_i(m \otimes a_1 \otimes \cdots \otimes a_n) = m \otimes a_1 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{n-1}$ for $1 \leq i \leq n$.
- $d_n(m \otimes a_1 \otimes \cdots \otimes a_n) = a_n m \otimes a_1 \otimes \cdots \otimes a_{n-1}$.

The Hochschild complex comes from the associated chain complex of the simplicial object defined as follows.

The construction of the Hochschild complex parallels that of the bar construction. Indeed, we can relate them via the following.

Definition 3.2.2 (Hochschild Homology)

Let R be a commutative ring. Let A be an R -algebra. Let M be an (A, A) -bimodule. Define the Hochschild homology of M to be

$$HH_n(A; M) = H_n(C_*(A, M))$$

Proposition 3.2.3

Let R be a commutative ring. Let A be an R -algebra. Let M be an (A, A) -bimodule. Then there is an isomorphism of chain complexes

$$C_*(A, M) \cong M \otimes_{A^e} B_*(A, A)$$

Motivated by the above relation with the bar complex, we define Hochschild cohomology as follows.

Definition 3.2.4 (The Hochschild Cocomplex)

Let R be a commutative ring. Let A be an R -algebra. Let M be an (A, A) -bimodule. Define the Hochschild cocomplex of M to be

$$C^*(A, M) = \text{Hom}_R(A^{\otimes_R^n}, M)$$

Definition 3.2.5 (Hochschild Cohomology)

Let R be a commutative ring. Let A be an R -algebra. Let M be an (A, A) -bimodule. Define the Hochschild cohomology of M to be

$$HH^n(A; M) = H^n(C^*(A, M))$$

Proposition 3.2.6

Let R be a commutative ring. Let A be an R -algebra. Let M be an (A, A) -bimodule. Then there is an isomorphism of chain complexes

$$C^*(A, M) \cong \text{Hom}_{A^e}(B_*(A, A), M)$$

Lemma 3.2.7

Let G be a group. Let M be a right G -module. Then there are natural isomorphisms

$$H_*(G; M) \cong HH_*(\mathbb{Z}[G]; M) \quad \text{and} \quad H^*(G; M) \cong HH^*(\mathbb{Z}[G]; M)$$

where M is considered as a (G, G) -bimodule with the trivial action on the left.

3.3 Relation to Tor and Ext**Proposition 3.3.1**

Let R be a commutative ring. Let A be an R -algebra. Let M be an (A, A) -bimodule. If A is flat as an R -module, then there is an isomorphism

$$HH_*(A, M) \cong \text{Tor}_*^{R^e}(A, M)$$

Proposition 3.3.2

Let R be a commutative ring. Let A be an R -algebra. Let M be an (A, A) -bimodule. If A is projective as an R -module, then there is an isomorphism

$$HH^*(A, M) \cong \text{Ext}_{R^e}^*(A, M)$$

Indeed, the tensor product is adjoint to the hom functor. The type of objects that allow for the tensor product and the hom functor to be exact would be the flat and projective objects respectively.

Cr1 of the above: If R is a field, then explicit description via absolute tor and ext.

Explicit description via relative tor and ext

3.4 Low Degree Interpretations**Proposition 3.4.1**

Let R be a commutative ring. Let A be an R -algebra. Let M be an (A, A) -bimodule. Then we have

$$HH_0(A, M) \cong \frac{M}{k\langle[A, M]\rangle} \quad \text{and} \quad HH^0(A, M) \cong \{m \in M \mid am = ma \text{ for all } a \in A\}$$

Proposition 3.4.2

Let R be a commutative ring. Let A be an R -algebra. Let M be an (A, A) -bimodule. Then we have

$$HH_1(A, M) \cong \Omega_{A/R}^1 \otimes_R M \quad \text{and} \quad HH^1(A, M) \cong \frac{\text{Der}_R(A, M)}{\text{Inn}_R(A, M)}$$

3.5 The Trace Map

Definition 3.5.1 (The Generalized Trace Map) Let R be a ring and let M be an R -module. Define the generalized trace map

$$\text{tr} : M_r(M) \otimes M_r(A)^{\oplus n} \rightarrow M \otimes A^{\otimes n}$$

by the formula

$$\mathrm{tr}((m_{i,j}) \otimes (a_{i,j})_1 \otimes \cdots \otimes (a_{i,j})_n) = \sum_{0 \leq i_0, \dots, i_n \leq r} m_{i_0, i_1} \otimes (a_{i_1, i_2})_1 \otimes \cdots \otimes (a_{i_n, i_0})_n$$

Theorem 3.5.2 The trace map defines a morphism of chain complex

$$\mathrm{tr} : C_{\bullet}(M_r(A), M_r(M)) \rightarrow C_{\bullet}(A, M)$$

3.6 Morita Equivalence and Morita Invariance

Definition 3.6.1 Let R and S be rings. We say that R and S are Morita equivalent if there is an equivalence of categories

$$\mathrm{Mod}_R \cong \mathrm{Mod}_S$$

Theorem 3.6.2 (Morita Invariance for Matrices)

4 (Co)Homology for Lie Algebras